Mountains vs. Beaches Vacation Preference Classification

Predicting vacation preference (beach or mountain) from demographic and lifestyle features using three classifier families with cross-validation and overfitting analysis.

Jibril Hussaini | Python · scikit-learn · pandas · NumPy · matplotlib · seaborn

Overview

This project builds and compares classifiers that predict whether a person prefers a beach or a mountain vacation, based on 13 demographic and lifestyle features drawn from a 52,444-row Kaggle dataset. The classes are imbalanced (roughly 75% beach, 25% mountain), so accuracy is reported alongside precision, recall, and F1 for the minority class. Three model families are evaluated: a decision tree, k-nearest neighbours, and a multi-layer perceptron.

Goals

Deliver a well-validated preference classifier, identify the features that drive the prediction most strongly, and demonstrate how overfitting can be detected and corrected in a decision tree without sacrificing accuracy.

Objectives

- Compare a Decision Tree, KNN, and a Neural Network (MLP) using 10-fold stratified cross-validation and a held-out test set.
- Study overfitting in the decision tree by plotting training and cross-validation accuracy against tree depth, then apply depth limiting to resolve it.
- Tune KNN with a grid search over the number of neighbours and the distance-weighting scheme to close the gap with the other models.
- Use the decision tree feature importances to identify the variables that drive vacation preference.
- Report accuracy, precision, recall, and F1 for the mountain class on the held-out test set for every model.

Approach

Categorical columns are one-hot encoded and numeric columns are standardised inside a ColumnTransformer fitted on the training split only. Each model is wrapped in a pipeline with that preprocessor and assessed with 10-fold stratified cross-validation before a final evaluation on the held-out 20% test set. The decision tree is also trained across depths 1 through 20 to produce an overfitting curve, and KNN is tuned via grid search over k from 1 to 30 and uniform versus distance weighting.

Outcomes

- The Neural Network (MLP) reached 0.998 test accuracy and 0.996 F1 on the mountain class.
- The Decision Tree matched the network at 0.996 test accuracy and 0.992 F1 with a depth limit of 10, eliminating the overfitting seen in the unconstrained tree.
- Tuned KNN (15 neighbours, distance weighting) improved test F1 from 0.831 to 0.865 and accuracy from 0.920 to 0.939.

- Proximity to mountains, proximity to beaches, and preferred activity together account for roughly two-thirds of the decision tree feature importance, explaining why even a shallow tree classifies nearly perfectly.